

GOKULANATH K

Software Developer | Computer Vision & Assistive Tech Specialist

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PROFESSIONAL SUMMARY

Resourceful Software Developer and MCA Candidate specializing in Computer Vision, Human-Computer Interaction (HCI), and Assistive Technology. Published patent co-inventor, international researcher (CAIT 2026 - HICAS), and Smart India Hackathon award winner with core expertise in Python, Flask, OpenCV, MediaPipe, Scikit-Learn, SQL, and Full-Stack Web Architecture.

TECHNICAL SKILLS

Languages & Databases: Python, Java, C++, SQL (MySQL, SQLite), JavaScript (ES6+), HTML5, CSS3
Computer Vision & AI: OpenCV, MediaPipe, Pose Estimation, Scikit-Learn, Facial Recognition, EAR Algorithm
Web & Developer Tools: Flask, React.js, Node.js, Express, RESTful APIs, Telegram Bot API, Docker, Git/GitHub
Core Competencies: Digital Rehabilitation Tech, Hands-Hands HCI, Machine Learning, Agile Collaboration

KEY PROJECTS

RehabCare – AI-Powered Digital Rehabilitation Platform 2026

Technologies: Python, Flask, OpenCV, MediaPipe, Scikit-Learn, Telegram Bot API

- Architected an end-to-end digital rehabilitation suite scoring patient exercise technique and counting repetitions in real-time via live webcams or video uploads using MediaPipe pose estimation.
- Engineered dual hands-free communication channels (hand-gesture recognition & eye-blink Morse code) enabling speech/motor-impaired patients to interact directly with medical care teams.
- Trained machine learning models to dynamically recommend next-session exercise difficulty based on historical recovery metrics, integrated with a live Telegram bot for real-time doctor alerts.

Customized Eye-Blink Assistive Communication System for Motor-Impaired Patients 2025 – 2026

Technologies: Python, OpenCV, Eye Aspect Ratio (EAR) Algorithm, Telepresence Integration

- Engineered a low-cost, hands-free assistive communication system utilizing webcam-based real-time eye-blink tracking to convert blink patterns into emergency alerts and messages.
- Integrated doctor interaction logging and telepresence alerting, significantly reducing patient response time and setup costs compared to expensive commercial alternatives.

Real-Time Hand Gesture-Based Mouse & Computer Control 2025

Technologies: Python, OpenCV, MediaPipe, PyAutoGUI

- Developed a touchless computer control interface tracking 21 hand landmarks via webcam to map hand movements directly to OS cursor movement, clicks, and window controls.

Face Recognition with Dynamic User Registration 2024 – 2025

Technologies: Python, OpenCV, Deep Learning Vector Embeddings, SQLite

- Built an intelligent face recognition pipeline that identifies users in live video feeds and registers new profiles on-the-fly without requiring full model retraining or system downtime.

PATENTS & PUBLICATIONS

- Published Indian Patent Application (Co-Inventor):** "An Adaptive Multimodal Communication and Classroom Participation System for Students with Disabilities" (App No: **202641107831 A**, Published: Sept 2026). Developed software-based multimodal accessibility integration using gesture tracking, eye-gaze navigation, and customizable eye-blink Morse code.
- Published International Research Paper:** Authored "Customized Eye Blink Based Communication System for Paralysed Person" published in the **7th International Conference on Computer Applications and Information Technology (CAIT 2026) (HICAS)**.
- Best Idea Award:** Awarded Best Idea Award in the **SIH Internal Hackathon 2025**.

EDUCATION

Master of Computer Applications (MCA) — CGPA: 7.9 2025 – Present

Hindusthan College of Arts and Science, Coimbatore, TN

Bachelor of Computer Applications (BCA) 2022 – 2025

Hindusthan College of Arts and Science, Coimbatore, TN

Higher Secondary (12th Grade) 2021 – 2022

NM Matriculation Higher Secondary School